SOCIAL NETWORKS · PAPER PRESENTATION



Predicting, evaluating, and explaining top misinformation spreaders via archetypal user behavior

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BEFORE WE START · COVID-19, APRIL 2020

“I see the disinfectant, where it knocks it out in a minute... And is there a way we can do something like that, by injection inside or almost a cleaning.”

— Donald Trump, White House coronavirus briefing, 23 April 2020

Next day, the maker of Lysol and Dettol warned that its disinfectants must never be put into the human body.

One high-reach account can push a claim to millions in hours. Can we spot such accounts before they do?

MOTIVATION

Moderation today arrives after the damage is done

01 A misleading post is published

02 It is reshared and spreads widely

03 Users, filters or fact-checkers flag it

04 Labeled, demoted or removed

> **By step 4** it has already reached large audiences — false news spreads faster and wider than corrections.

RESEARCH QUESTION

“Does an account’s behavioral footprint reveal latent signals of misinformation risk?”

Goal: **rank users** by their potential to spread misinformation, so platforms can act **before** content goes viral.

METHOD OVERVIEW

The pipeline in six steps

01 Collect

COVID-19 reshares from Twitter: VaccinItaly and Multilanguage

02 Label

Rate each URL's domain with NewsGuard; ≤ 39 = low-credibility

03 Score

Six archetype features per user from the observation period

04 Rank

One score per archetype, or a Random Forest combining all six

05 Evaluate

Dismantle the future reshare network: Quality@k and nDCG@k

06 Explain

SHAP shows which archetype drives each user's rank

Content-agnostic: only a credibility label is needed, so the same pipeline works for images or video.

DATA

Two COVID-19 Twitter datasets, labeled by source credibility

Credibility labeling

Each URL's domain gets a **NewsGuard** trust score from 0 to 100.

Score **≤ 39 → low-credibility** (misinformation).

Only original posts and reshares count — no replies, quotes or self-reshares.

VaccinItaly

Italian · Dec 2020 – Oct 2021 · 306 days

371,586

retweets

54,916 original posts

51,962 users

COVID-19 Multilanguage

Global · Nov 2020 – Nov 2021 · 372 days

1,118,697

reshares

103,250 original posts

129,507 users

BEHAVIORAL ARCHETYPES

Three ways an account can drive misinformation

Amplifiers

Create little original content but boost others' posts through extensive resharing.

Signal: how often they reshare low-credibility posts

Coordinated accounts

Groups acting in concert to push the same narratives, simulating consensus.

Signal: synchronized, highly similar resharing behavior

Super-spreaders

Post original content that consistently goes viral and attracts heavy engagement.

Signal: volume and virality of reshares they receive

Archetypes are **not mutually exclusive** — one account can show several — and they cover humans and bots alike, since only behavior is used.

RANKING METHODS

One family of ranking scores per archetype

Archetype	Score	How users are ranked
Amplifiers	Repost Count	Number of low-credibility posts the user reshared
	Early Reposter (EaR) Index	Rewards resharing early in a cascade, especially of later-viral posts
Coordinated	Coord-Centrality	Eigenvector centrality in the co-reshare similarity network
	Coord-EdgeWeight	Strongest cosine-similarity link to any other user
Super-spreaders	Influence Score	Total reshares received by the user's low-credibility posts
	H-Index	Largest h such that h posts each got at least h reshares
	TAI-Score · TASH-Index	New: time-aware versions of the two scores above

KEY CONTRIBUTION

Time-aware scores: TAI-Score and TASH-Index

TAI-SCORE — TIME-AWARE INFLUENCE SCORE

$$TAI(u)_t = \alpha \cdot TAI(u)_{t-1} + (1 - \alpha) \cdot Influence(u)_t$$

TASH-INDEX — TIME-AWARE SOCIAL H-INDEX

$$TASH(u)_t = \alpha \cdot TASH(u)_{t-1} + (1 - \alpha) \cdot H(u)_t$$

Split time into slots of δ days, compute the score in each slot, smooth with an exponential moving average. The first slot's value starts the average.

Why time-aware?

A fixed window is arbitrary and biased by posting frequency.

The α knob

α = 1: only the first slot. α = 0: only the latest slot.

Grid search on nDCG

TASH: α = 0.5, δ = 14 days

TAI: α = 0.6, δ = 18 days

MACHINE LEARNING RANKING

Combining all archetypes in one learning-to-rank model

1 · INPUT FEATURES

Repost Count · EaR-Index

Coord-Centrality · Coord-EdgeWeight

TAI-Score · TASH-Index

→

2 · MODEL

Random Forest regression — pointwise ranking (best of several off-the-shelf models)

→

3 · TARGET

User **strength**: total low-credibility reshares the user is involved in, as author or resharer

→

4 · EXPLAIN

SHAP shows which archetype drives each rank

Temporal split, no leakage: the first 80% of reshares (in time order) are the observation period for training; the last 20% are the evaluation period.

EVALUATION

Evaluation by dismantling the misinformation network

Reshare network from the test period: edge A → B weighted by how many of A's low-credibility posts B reshared.

Remove users in ranked order and measure how much misinformation volume disappears.

> **Quality@k** — share of misinformation removed by moderating the top-k users

> **nDCG@k** — agreement with the ground-truth ranking by node strength

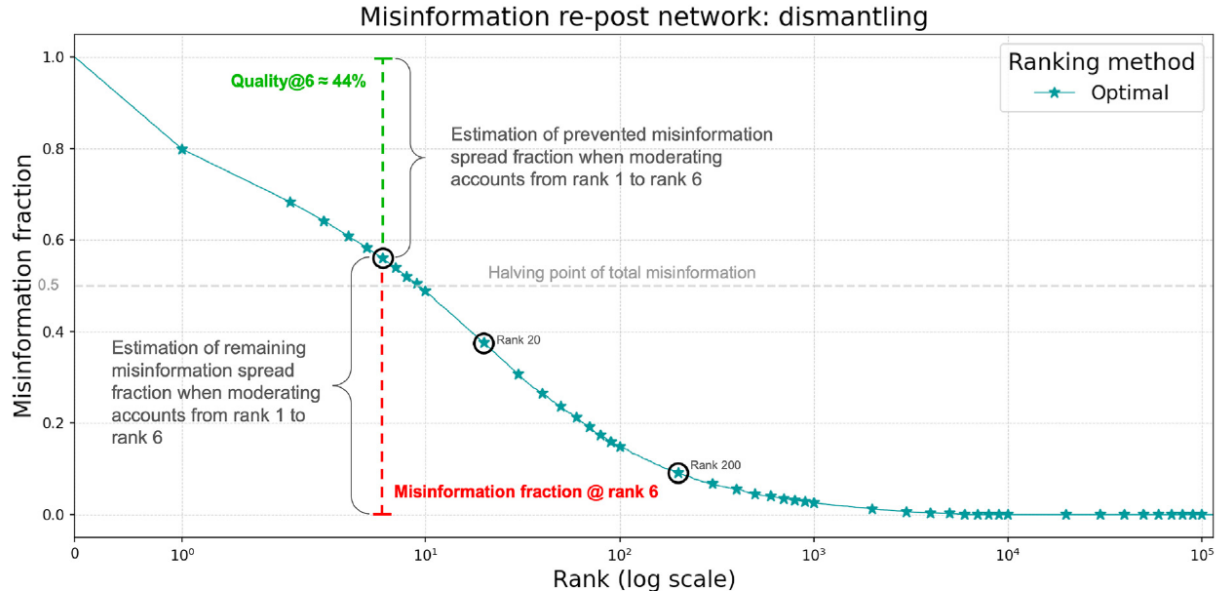
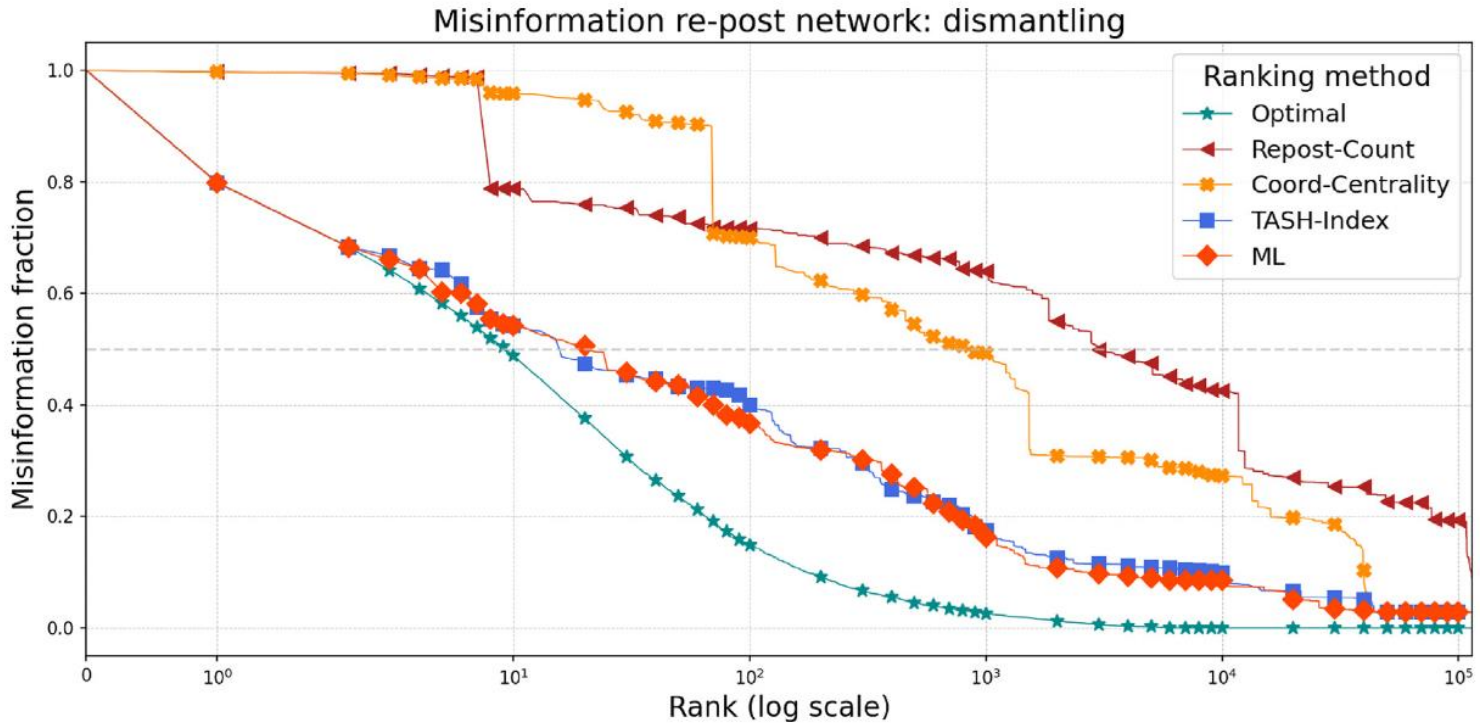


Fig. 2 — optimal ranking: removing just the top 6 users cuts ≈ 44% of misinformation.

RESULTS · RANKING PERFORMANCE

ML and TASH-Index track the optimal curve



Lower curve = better. Each point is one more user removed.

ML and **TASH-Index** stay close to optimal for top ranks.

Amplifier and coordination scores alone lag far behind.

Fig. 3 · VaccinItaly dataset

RESULTS · EXPLAINABILITY

What drives the model overall: SHAP summary

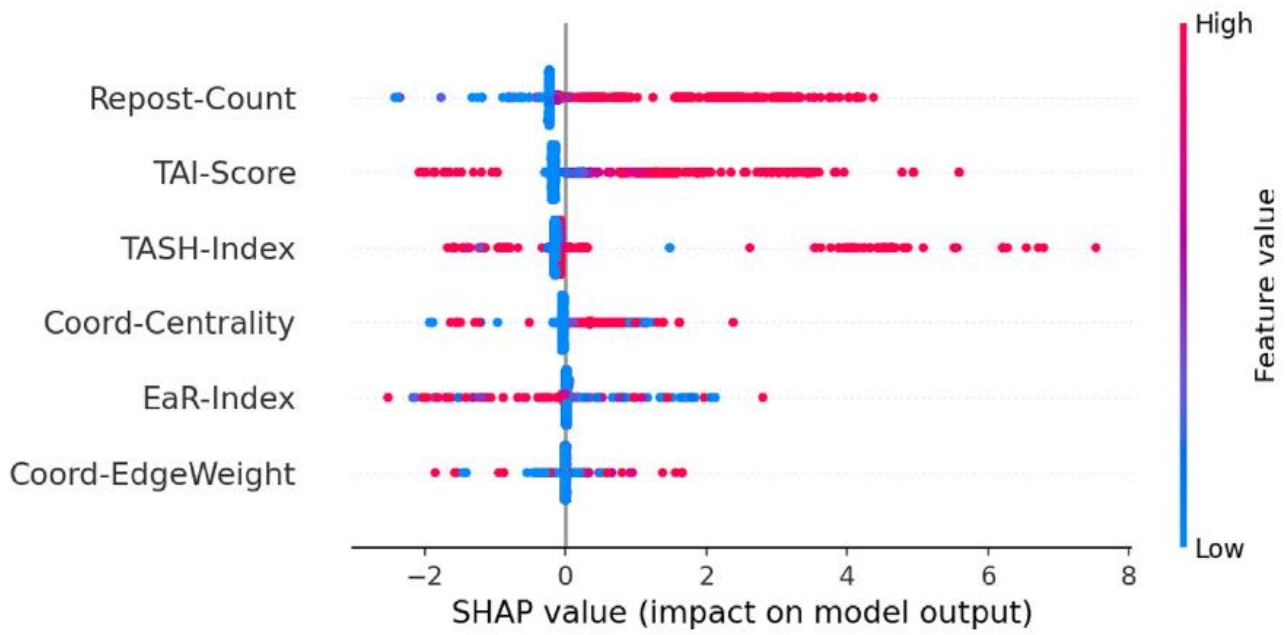


Fig. 4 — one dot per user · red = high feature value · right of 0 = pushes the user up the ranking

#1

Repost Count (amplifiers): frequent resharing is the strongest signal

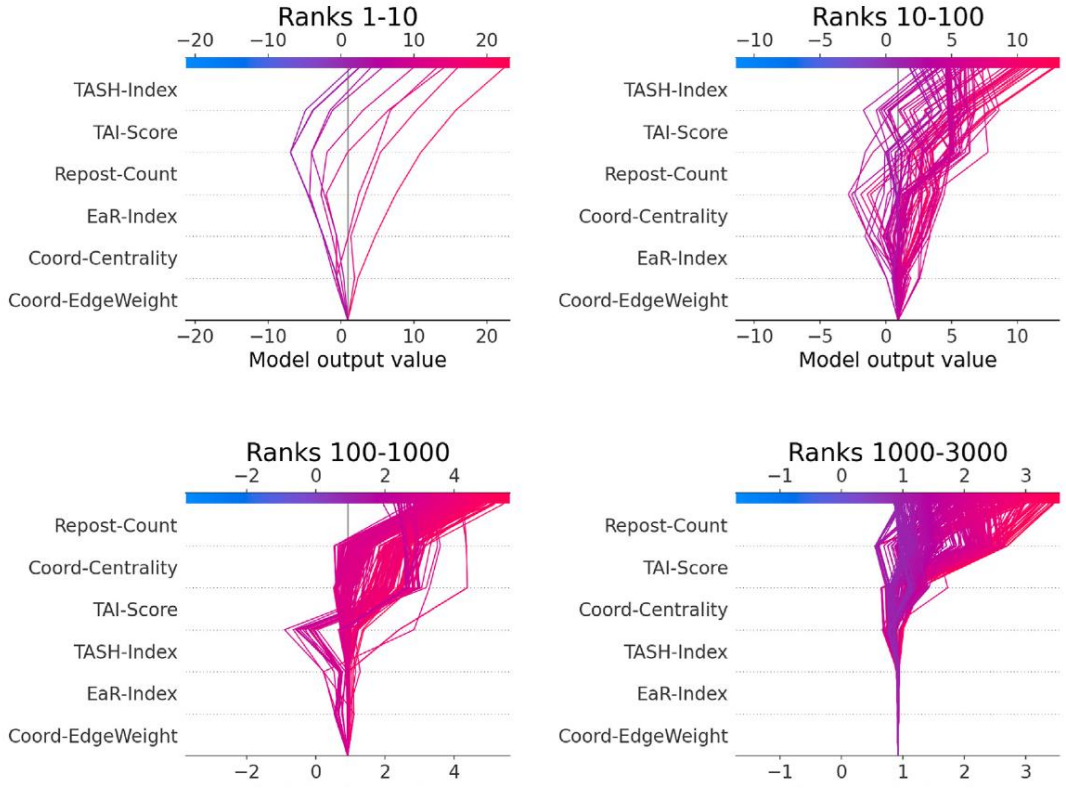
#2–3

TAI-Score and **TASH-Index** (super-spreaders)

Coordination and EaR features contribute less.

RESULTS · EXPLAINABILITY BY RANK

Different archetypes matter at different ranks



1–10	TASH & TAI lead → **super-spreaders**
10–100	Repost Count and Coord-Centrality join → **hybrid** users
100–1000	Repost Count leads → **amplifiers**, with **coordination** rising
1000–3000	Repost Count still first → the long tail

Super-spreaders start the cascade; amplifiers and coordinated accounts **keep it going**.

Fig. 5 · SHAP decision plots per rank interval

RESULTS · DATA SCARCITY

How much history does each model need?

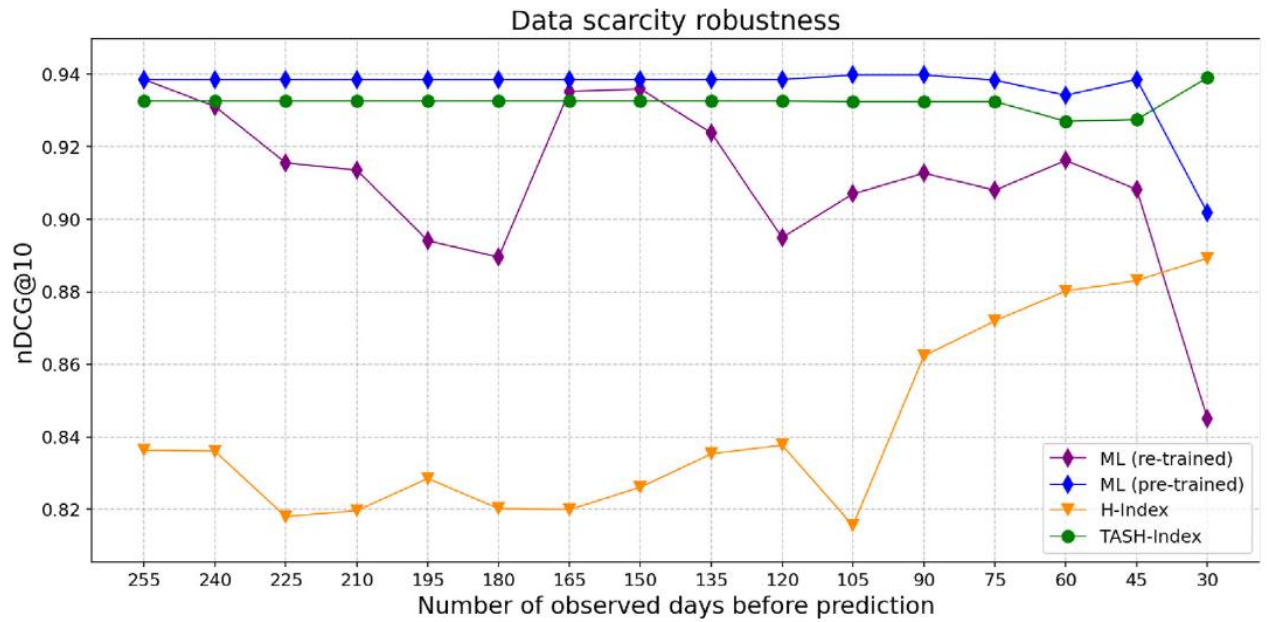


Fig. 6 — nDCG@10 as the observation window shrinks from 255 to 30 days.

TASH-Index

Stable at ≈ 0.93 even with 30 days

ML, pre-trained

Stable; drops only at 30 days

ML, re-trained

Unstable as data shrinks

H-Index

Lowest throughout

Less data needed → earlier intervention

IMPLICATIONS

From rankings to moderation: one action per tier

TOP RANKS **Super-spreaders**	**Targeted, preventive actions** Closer moderation scrutiny, fact-checking overlays, temporary content restrictions
MID AND LOWER RANKS **Amplifiers and coordinated accounts**	**Systemic measures** Limit amplification of their posts, reduce recommendation exposure, add friction: share limits, warning prompts

Decision support, not automatic bans: humans stay in the loop, with auditability and a right to appeal.

DISCUSSION

Limitations, our critique and future work

Limitations

- Only direct reshares; who actually saw a post is unknown
- Credibility judged per domain, not per post
- New users can’t be ranked: 2.9% (VaccinItaly) and 4.8% (Multilanguage) of misinformation is never removable
- Static dismantling ignores how users react to moderation

Our critique

- Intent is ignored: careless sharers and campaigners look alike
- Users missing from the test period count as zero harm
- A pre-trained model may drift as behavior changes
- Twitter and COVID-19 only; other platforms and topics untested

Future work

- Integrate TASH into graph neural networks
- Adaptive tuning of α and δ, e.g. Bayesian optimization
- Multi-platform data; inductive models for new users
- Simulations of interventions’ long-term effects

TAKEAWAYS

What we learn from this paper

1. **A few users drive most misinformation.** Super-spreaders dominate the top ranks; amplifiers and coordinated accounts sustain the spread further down.

2. **Time matters.** The simple, interpretable TASH-Index nearly matches ML and stays robust with little data.

3. **Explainable rankings support human moderators** — closer review for top spreaders, friction such as share limits for amplifiers.

Thank you

Questions?

